

# L4YAML: A Verified YAML 1.2.2 Parser in Lean 4

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L4YAML is a fully verified YAML 1.2.2 parser written in pure Lean 4. It delivers 2,309 machine-checked theorems across 61 proof modules, zero axioms, zero `sorry`, and zero `partial def` — while passing 100% of the YAML 1.2.2 specification examples and 100% of the applicable `yaml-test-suite` test IDs (225/225).

This manual documents the project's architecture, verification strategy, security model, and FFI bindings for C, Python, and Rust.

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- 1. Overview
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## 2. Architecture

L4YAML implements a two-pass pipeline that mirrors the layered structure of the YAML 1.2.2 specification itself. Input bytes flow through a character-level [scanner](#), producing a typed token stream, which a [recursive-descent parser](#) converts into a structured AST.

### 2.1. Scanner (Lexical Layer)

### 2.2. Token Parser (Syntactic Layer)

### 2.3. Type System

### 2.4. Module Organization

### 2.5. Import Graph



## 2.5. Import Graph

The project's dependency structure can be visualized using `lake exe graph`. The FFI layer sits at the top, depending on `Config`, which in turn depends on the scanner/parser/schema pipeline:

```
FFI ← Config ← Dump ← Schema ← Token ← Scanner ← TokenParser
      ↑
CharPredicates ← YamlSpec
```

The full import graph and component-level dependency graphs are available in the `graphs/` directory as Graphviz DOT files.



## 2.4. Module Organization

The project is organized into several module groups:

| Group      | Key Modules                                                              | Purpose                                                                         |
|------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Core       | Types.lean, Token.lean, Grammar.lean, YamlSpec.lean, CharPredicates.lean | Type definitions, token types, grammar inductive, spec production predicates    |
| Scanner    | Scanner.lean, Scanner/Whitespace.lean, Scanner/Scalar.lean, ...          | Character-to-token conversion with full state management                        |
| Parser     | TokenParser.lean, NodeParser.lean                                        | Token-to-AST recursive descent                                                  |
| Validation | Limits.lean, Schema.lean                                                 | Security limits, Core Schema type resolution                                    |
| Output     | Emitter.lean, Dump.lean, RoundTrip.lean                                  | Canonical emitter (~164L), style-aware dump, round-trip properties              |
| FFI        | FFI.lean, ffi/, python/, rust/                                           | C/Python/Rust bindings via @[export]                                            |
| Proofs     | Proofs/ (61 modules, ~47,000 lines)                                      | Machine-checked theorems for soundness, completeness, progress, well-formedness |





## 2.1. Scanner (Lexical Layer)

The scanner (`Scanner.lean`, ~920 lines) converts a UTF-8 input string into a stream of `YamlToken` values. It implements 132 YAML productions from the specification and maintains several pieces of state:

- *Indentation stack* — tracks block-level nesting via column positions
- *Flow level counter* — distinguishes block context from flow context (`[ ]`, `{ }`)
- *Simple key state* — manages the YAML "simple key" lifecycle where a plain scalar may retroactively become a mapping key when `:` is encountered
- *Anchor map* — records anchor names for alias resolution
- *Position tracking* — offset, line, and column for every token

The scanner returns `Except ScanError (Array YamlToken)`, where `ScanError` carries position information and a human-readable message.

### 2.1.1. Token Types

The `YamlToken` inductive defines the full set of lexical elements:

- Stream markers: `streamStart`, `streamEnd`
- Document markers: `documentStart`, `documentEnd`
- Structure indicators: `blockSequenceStart`, `blockMappingStart`, `blockEnd`, `flowSequenceStart`, `flowSequenceEnd`, `flowMappingStart`, `flowMappingEnd`, `key`, `value`, `blockEntry`, `flowEntry`
- Content tokens: `scalar` (with style tag: `plain`, `single-quoted`, `double-quoted`, `literal`, `folded`), `anchor`, `alias`, `tag`
- Placeholder tokens: used as reservation slots for simple key backpatching (filtered before output)

### 2.1.2. Append-Only Design

A critical design choice is the *append-only token stream*. When the scanner encounters a potential simple key, it pushes two placeholder tokens (for the future key and `blockMappingStart` indicators) and records their indices. If the key is later confirmed (by encountering `:`), these placeholders are overwritten in-place via `setIfInBounds`. If not confirmed, they remain as placeholders and are filtered out before the token stream is returned.

This avoids `insertAt` operations that would shift array indices and invalidate saved positions — a property that is essential for the formal proof that the scanner makes monotonic progress.



## 2.2. Token Parser (Syntactic Layer)

The token parser (`TokenParser.lean`, ~426 lines) consumes the `YamlToken` stream and produces an `Array YamlDocument`. It implements 54 YAML productions via hand-written recursive descent (no parser combinator library).

Key responsibilities:

- *Multi-document support* — handles `---` and `...` boundaries
- *Alias resolution* — substitutes `*anchor` references with previously anchored values
- *Tag resolution* — applies `%TAG` directive handle expansion and schema-level type resolution
- *Schema application* — resolves untagged scalars to typed values (`null`, `bool`, `int`, `float`, `str`) via the Core Schema



## 2.3. Type System

The output AST is centered on `YamlValue`, a compact inductive type:

- `YamlValue.scalar` — tagged string value with resolved `YamlType`
- `YamlValue.sequence` — ordered list of values
- `YamlValue.mapping` — list of key-value pairs (preserving order)

Each value is wrapped in `YamlDocument`, which carries document-level metadata including directive tags and version indicators.

Position tracking is provided by `YamlPos` (offset, line, column), enabling precise error reporting that maps back to the original input.



# 5. FFI Bindings

L4YAML provides foreign function interface bindings for C, Python, and Rust, enabling the verified parser to be used from mainstream languages. The FFI layer sits above the full parser pipeline — callers get the same verified parsing logic, security limits, and schema resolution available in Lean.

## 5.1. Dependency Graph

## 5.2. C API

## 5.3. Python API

## 5.4. Rust API

## 5.5. Adding FFI Bindings





## 5.5. Adding FFI Bindings

To expose a new Lean function via FFI:

1. Add `@[export l4yaml_function_name]` to the Lean definition
2. Declare the corresponding C prototype in `ffi/l4yaml.h`
3. Add any necessary shim code in `ffi/l4yaml_shim.c`
4. Update the Python/Rust wrappers as needed
5. Add tests at each layer

The `@[export]` attribute instructs the Lean compiler to generate a C-callable symbol with the specified name.



## 5.2. C API

The C API is defined in `ffi/l4yaml.h` with a bridge layer in `ffi/l4yaml_shim.c`. It provides approximately 26 exported functions covering:

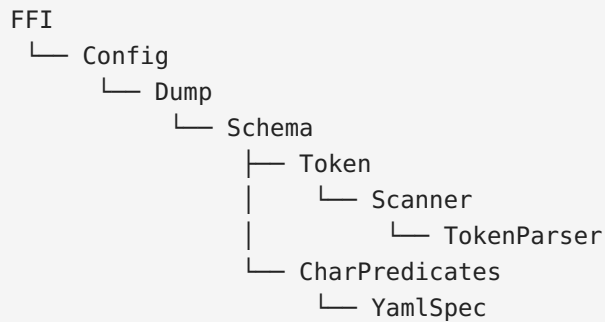
- *Parsing* — `l4yaml_parse_safe` and related functions
- *Dumping* — converting `YamlValue` back to YAML text
- *Schema* — tag handle registration and type resolution
- *Limits* — configuring `ParserLimits` via preset codes or individual parameter setters
- *Memory* — proper Lean object lifecycle management (`lean_inc_ref` / `lean_dec_ref`)

All C API functions operate on opaque Lean object pointers. The shim layer handles the Lean runtime initialization and provides stable C-callable entry points.



# 5.1. Dependency Graph

The FFI module sits at the top of the dependency chain:



`FFI.lean` exports functions via `@[export]` that are callable from C. The Python and Rust bindings wrap these C-level exports with idiomatic APIs in their respective languages.



## 5.3. Python API

The Python package (`python/l4yaml/`) wraps the C API using `ctypes`, providing a Pythonic interface with:

- Full type annotations for IDE support
- Automatic garbage collection of Lean objects via reference counting
- Cross-platform support (Linux, macOS, Windows)
- 78 tests covering the full API surface

Usage follows the familiar pattern:

```
from l4yaml import parse

result = parse("key: value")
```





## 5.4. Rust API

The Rust workspace (`rust/`) provides two crates:

- **`l4yaml-sys`** — raw FFI bindings generated from the C header. Unsafe, low-level, intended for direct interop.
- **`l4yaml`** — safe RAI wrapper with 21 tests. Lean object lifetimes are managed via the `Drop` trait, ensuring proper cleanup without manual reference counting.

The safe wrapper provides idiomatic Rust types and error handling via `Result`, translating Lean's `Except` into Rust's error model.



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# 1. Overview

L4YAML is a pure Lean 4 implementation of the YAML 1.2.2 specification (RFC 9512). No external parsing libraries, no C dependencies in the core, and no use of `partial def` — every function is provably terminating.

The project demonstrates that a production-quality parser for a complex real-world format can be built and formally verified in Lean 4, with practical performance and comprehensive test coverage.

## 1.1. At a Glance

## 1.2. Key Design Decisions



# 1.1. At a Glance

| Key Metric                   | Value                                         |
|------------------------------|-----------------------------------------------|
| Machine-checked theorems     | 2,309 across 61 proof modules (~47,000 lines) |
| Compile-time #guard tests    | 2,124 (kernel-evaluated at build time)        |
| Axioms / sorry / partial def | Zero / Zero / Zero                            |
| Build jobs                   | 415/415 (clean, zero warnings)                |
| Runtime test suites          | 1,041 tests across 19 suites                  |
| Spec examples passing        | 132/132 (100%)                                |
| yaml-test-suite IDs          | 225/225 YAML 1.2.2-applicable (100%)          |
| yaml-test-suite total        | 354/406 (87.2%; 52 skipped are YAML 1.1/1.3)  |



## 1.2. Key Design Decisions

The parser is built around several deliberate design choices:

- **Pure Lean 4, zero external dependencies.** The core parser has no C FFI calls and no dependency on external parsing libraries. This ensures every line of parsing logic is visible to the Lean kernel for formal verification.
- **Total functions only.** Every function terminates provably. The `partial` keyword is never used in production code. Termination is proved via well-founded recursion over input offsets, indentation levels, and flow nesting depth.
- **Append-only token streams.** Tokens are emitted into a pre-allocated array with placeholder reservation slots. Backpatching uses `setIfInBounds` (bounded update) rather than `insertAt` (which would shift indices). This design ensures monotonic progress and simplifies the formal proof of scanner correctness.
- **Two-pass architecture.** A character-level scanner emits a `YamlToken` stream, which a separate recursive-descent parser converts to a typed AST. This separation mirrors the YAML specification's own layered structure and enables independent verification of each layer.
- **Configurable security limits.** All resource bounds (nesting depth, string length, collection sizes) are configurable via `ParserLimits`, with four built-in presets from `strict` to `unlimited`.



# 4. Security

YAML parsing is a well-known attack surface. L4YAML addresses this with configurable parser limits, preset security profiles, and formal verification of the parsing logic itself.

## 4.1. Threat Model

## 4.2. ParserLimits

## 4.3. Preset Configurations

## 4.4. Verification of Security Properties





## 4.2. ParserLimits

The ParserLimits structure provides 11 configurable threat mitigations:

| Parameter              | Default      | Purpose                                                        |
|------------------------|--------------|----------------------------------------------------------------|
| nestingDepth           | 250          | Maximum recursion depth — prevents billion-laugh expansion     |
| maxLength              | 10 MB        | Maximum scalar string length — DoS prevention                  |
| maxArrayLength         | 100,000      | Maximum sequence element count                                 |
| maxObjectSize          | 10 MB        | Maximum total mapping size                                     |
| maxAliasDepth          | 50           | Maximum alias chain depth — recursive cycle protection         |
| allowDuplicateKeys     | false        | Whether duplicate mapping keys are accepted                    |
| allowedTagHandles      | customizable | Restricts which tag handles (!, !!, custom) are permitted      |
| forbiddenTags          | customizable | Explicit rejection of dangerous tags (e.g., !!python/object)   |
| parseErrorPolicy       | strict       | Whether non-conformant input is rejected or best-effort parsed |
| commentEncoding        | explicit     | Character encoding validation for comments                     |
| literalNewlineHandling | standard     | Newline normalization in literal scalars                       |



## 4.3. Preset Configurations

Four preset configurations are provided for common use cases:

- **strict** — all protections enabled at conservative thresholds. Recommended for processing untrusted input (e.g., user uploads, network-received configuration).
- **default** — balanced settings suitable for most applications. Limits are generous enough for typical configuration files while still preventing resource exhaustion.
- **permissive** — reduced validation for trusted input. Useful when parsing known-good YAML from controlled sources.
- **unlimited** — all limits disabled. Explicitly dangerous; intended only for testing or for processing input that has already been validated externally.

The FFI layer exposes presets via `presetToLimits`, which maps a `UInt8` preset code to the corresponding `ParserLimits` configuration.



# 4.1. Threat Model

YAML parsers face several categories of attack:

- *Billion-laugh attacks* — deeply nested aliases that expand exponentially, exhausting memory
- *Denial of service* — extremely long strings, deeply nested structures, or very large collections that consume excessive resources
- *Arbitrary code execution* — YAML tags that trigger object deserialization in languages with unsafe constructors (e.g., Python's `!!python/object`)
- *Duplicate key confusion* — multiple identical keys in a mapping, where different consumers may pick different values

L4YAML mitigates all of these through its `ParserLimits` configuration.



## 4.4. Verification of Security Properties

The formal proofs establish that the parser correctly enforces its configured limits:

- Nesting depth is checked on every recursive call
- String length is checked during scalar accumulation
- Collection sizes are checked during sequence/mapping construction
- Alias depth is bounded during resolution

Because the parser is written in pure Lean 4 with no unsafe FFI in the core, there is no possibility of buffer overflows, use-after-free, or other memory safety violations in the parsing logic. The Lean runtime provides automatic memory management via reference-counted garbage collection.





# 7. Test Results

The L4YAML CI pipeline runs every test suite on each push and pull request. This chapter describes the test result categories and links to the most recent CI-generated reports.

Test results are generated by the `test-coverage.yml` GitHub Actions workflow and published alongside this documentation in the `reports/` directory.

The interactive coverage dashboard is available at [reports/index.html](https://reports/index.html).

## 7.1. yaml-test-suite Coverage

## 7.2. Lean Test Suites

## 7.3. Diagnostic Checks

## 7.4. FFI Test Results



## 7.3. Diagnostic Checks

Diagnostic pipelines produce detailed output for regression tracking:

| Diagnostic      | Output File             | Description                                |
|-----------------|-------------------------|--------------------------------------------|
| Flow Regression | flowregressioncheck.txt | Flow parsing edge case regression tracking |
| Error Stage     | errorstagediag.txt      | Error stage diagnostic output              |
| Scalar Stage    | scalarstagediag.txt     | Scalar stage diagnostic output             |



## 7.4. FFI Test Results

The FFI layer is tested through three independent paths:

### 7.4.1. Python FFI Tests

Two Python test suites validate the ctypes-based bindings:

- *FFI Integration Tests* (`Tests/test_python_ffi.py`) — tests the C shared library directly from Python
- *Package Tests* (`python/tests/`) — tests the `l4yaml` Python package API

Results are captured as both text logs and JUnit XML, with an HTML report generated from the JUnit data.

### 7.4.2. Cross-Language Comparison

The cross-language comparison runs the full `yaml-test-suite` through all four backends (Lean, C, Python, Rust) across five limit presets (`unlimited`, `default`, `strict`, `permissive`, `safe_tags`).

This verifies that the FFI layer preserves identical parsing behavior — every backend must produce the same pass/fail/skip results for every test case under every preset configuration.

Results are available at [Cross-Language Comparison Report](#).



## 7.2. Lean Test Suites

The following Lean-native test suites are executed and their output captured:

| Suite                 | Output File             | Description                               |
|-----------------------|-------------------------|-------------------------------------------|
| Unit Tests            | unit-tests.txt          | Core unit tests for parser internals      |
| Demo                  | demo.txt                | End-to-end parsing demonstrations         |
| Explicit Key Tests    | explicitkeytests.txt    | YAML explicit key (?) handling            |
| Flow Tests            | flowtests.txt           | Flow collection syntax completeness       |
| Validation Tests      | validationtests.txt     | Input validation and error reporting      |
| Dump Round-Trip       | dumproundtrip.txt       | Parse → dump → parse cycle verification   |
| Raw Parse Tests       | rawparsetests.txt       | Token-to-AST parsing validation           |
| Spec Examples         | specexamples.txt        | All 132 YAML 1.2.2 specification examples |
| Schema Dump           | schemadump.txt          | Schema resolution and dump formatting     |
| Scanner Tests         | scannertests.txt        | Scanner behavior on targeted inputs       |
| Scanner Spec Examples | scannerspecexamples.txt | Scanner-level tokens for spec examples    |
| Adversarial Tests     | adversarialtests.txt    | Handcrafted adversarial inputs            |
| Mutation Tests        | mutationtests.txt       | Systematic input perturbation             |
| Property Tests        | propertytests.txt       | Randomized property-based testing         |
| Production Coverage   | productioncoverage.txt  | Coverage across production-style inputs   |
| Limit Tests           | limittests.txt          | Parser security limit enforcement         |





# 7.1. yaml-test-suite Coverage

The `suiterunner` executable runs all 406 unique test cases from the official `yaml-test-suite` and generates interactive HTML coverage reports with filtering, sorting, and per-stage breakdown.

Interactive reports:

- [Full Coverage Report](#) — all test cases with filtering and sorting
- [Verified Tests Detail](#) — all internal test suites with per-test results
- [Production Coverage](#) — YAML 1.2.2 productions cross-referenced with `@[yaml_spec]` annotations

Coverage by stage:

- [Scalar Tests](#)
- [Flow Tests](#)
- [Block Tests](#)
- [Document Tests](#)
- [Advanced Tests](#)
- [Error Tests](#)



# 6. Testing

Beyond the 2,309 formal theorems, L4YAML maintains extensive runtime and compile-time test suites that validate the parser against real-world YAML inputs and the official specification.

## 6.1. Specification Coverage

## 6.2. Test Suites

## 6.3. Running Tests



## 6.3. Running Tests

All tests are built as Lake executables. To run the full default test suite:

```
lake build
```

This builds all 415 jobs (zero warnings) including all test executables listed in `defaultTargets`. Individual suites can be run directly:

```
lake exe specexamples  
lake exe scannertests  
lake exe adversarialtests
```

Test results are captured in `docs/` as both plain text and HTML reports.



## 6.1. Specification Coverage

- **132/132** YAML 1.2.2 specification examples pass (100%)
- **225/225** unique YAML test IDs from the yaml-test-suite pass (100% of YAML 1.2.2-applicable tests)
- **354/406** total yaml-test-suite tests pass (87.2%)
- The 52 skipped tests cover YAML 1.1 and YAML 1.3 features that are outside the YAML 1.2.2 scope





## 6.2. Test Suites

The project includes 24 test executables spanning 19 hand-written suites and 7 diagnostic pipelines:

| Suite               | Focus                                               |
|---------------------|-----------------------------------------------------|
| specexamples        | All 132 YAML 1.2.2 specification examples           |
| scannerspecexamples | Scanner-level token stream for spec examples        |
| scannertests        | Scanner behavior on targeted inputs                 |
| rawparsetests       | Raw token-to-AST parsing                            |
| flowtests           | Flow collection syntax (inline sequences, mappings) |
| flowregressioncheck | Regression tests for flow parsing edge cases        |
| explicitkeytests    | Explicit key (?) handling                           |
| validationtests     | Input validation and error reporting                |
| limittests          | Parser limit enforcement                            |
| adversarialtests    | Adversarial inputs targeting parser robustness      |
| mutationtests       | Systematic input perturbation                       |
| propertytests       | Randomized property-based testing                   |
| dumproundtrip       | Parse → dump → parse cycle validation               |
| schemadump          | Schema resolution and dump formatting               |
| productioncoverage  | Coverage analysis across production inputs          |
| errorstagediag      | Error stage diagnostic output                       |
| scalarestagediag    | Scalar stage diagnostic output                      |



# 3. Verification

L4YAML employs a three-layer verification strategy that combines formal proofs, compile-time checks, and runtime tests to achieve comprehensive coverage of the YAML 1.2.2 specification.

## 3.1. Three-Layer Strategy

## 3.2. Key Theorems

## 3.3. Key Proof Modules

## 3.4. Theorem Dependency Visualization

## 3.5. Proof Engineering Patterns

## 3.6. Zero-Axiom Policy



## 3.3. Key Proof Modules

| Module                  | Theorems                    | Scope                                                     |
|-------------------------|-----------------------------|-----------------------------------------------------------|
| ScannerCorrectness.lean | 259 theorems + 1,063 guards | Character-to-token correctness for all scanner operations |
| Completeness.lean       | 63 theorems                 | Valid YAML inputs are accepted                            |
| Soundness.lean          | 28 theorems                 | Output corresponds to valid grammar derivations           |
| RoundTrip.lean          | 58 theorems + 63 guards     | Parse-emit-parse cycle properties                         |
| Composition.lean        | 12 theorems                 | Scanner + parser pipeline correctness                     |
| ScannerEmitBridge.lean  | 12 theorems + 64 guards     | Bridge between scanner emissions and grammar predicates   |
| ParserSoundness.lean    | 12 theorems                 | Grammar-to-implementation correspondence                  |
| ParserWfaProofs.lean    | 50 theorems                 | Well-formed anchors through entire parser pipeline        |
| ParserNodeProofs.lean   | 57 theorems                 | Anchor growth and alias resolution                        |
| ParserWellBehaved.lean  | 102 theorems                | Flow nesting, token preservation, scannable properties    |
| ScannerProgress.lean    | 24 theorems                 | Offset strictly increases on every scanner step           |
| ScannerSimpleKey.lean   | Multiple theorems           | Simple key lifecycle well-formedness                      |
| ScannerDispatch.lean    | Multiple theorems           | Dispatch pipeline preserves all invariants                |



## 3.2. Key Theorems

The following capstone theorems represent the main formal guarantees established by L4YAML. Each is machine-checked by the Lean kernel with zero axioms beyond the built-in foundations.

Each theorem is accompanied by a bipartite dependency graph showing the implementation functions it proves properties about (blue, left) and the supporting theorems used in its proof (green, right). Stdlib lemmas surface in orange when they are domain-relevant.

### 3.2.1. Pipeline Composition

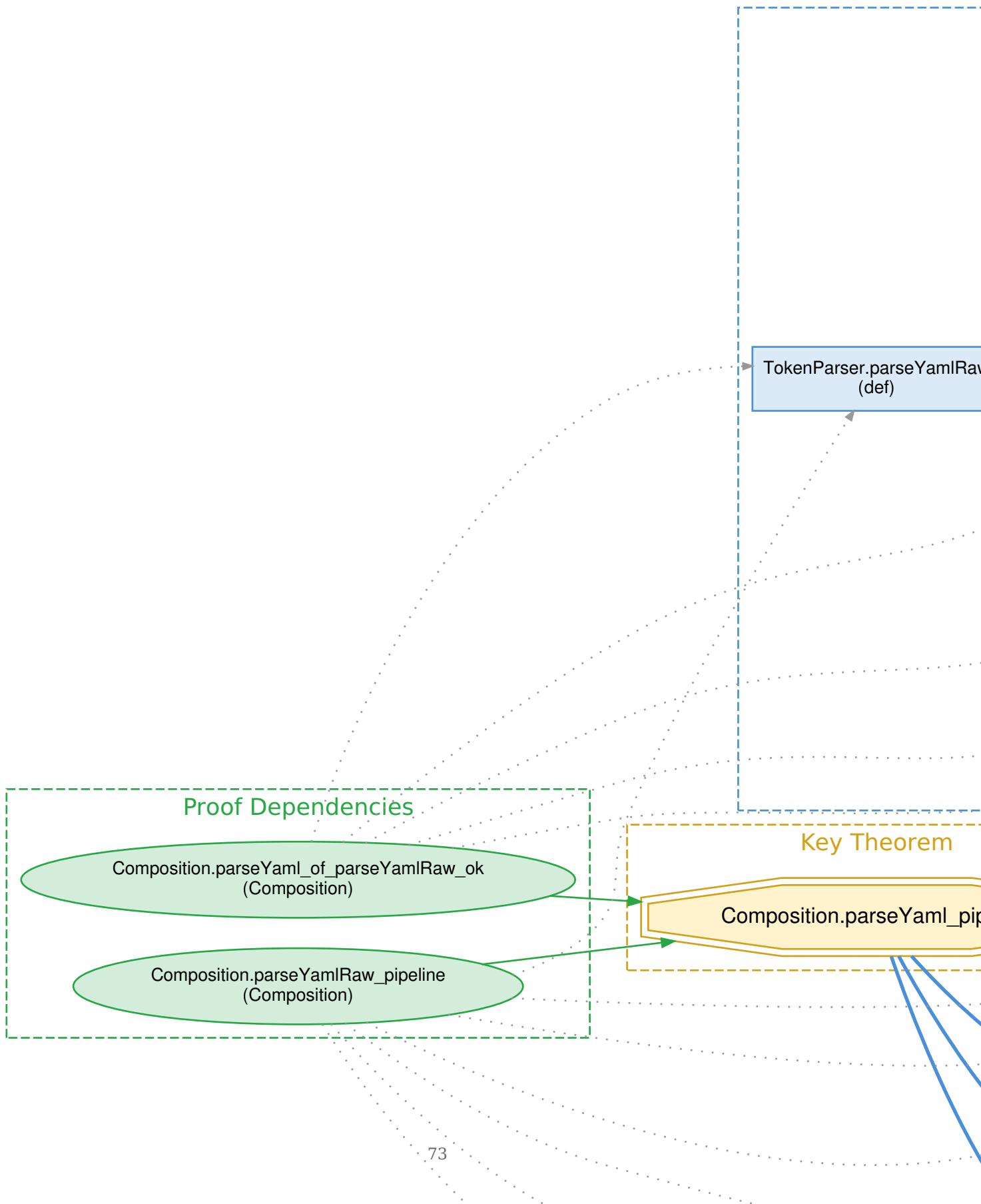
These theorems establish that the scanner and parser compose correctly into a single end-to-end pipeline.

| Theorem                                | Module       | Statement                                                                                                                                                                                          |
|----------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>parseYaml_pipeline</code>        | Composition  | End-to-end: scan then parse composes correctly. If <code>scanFiltered</code> produces tokens and <code>parseStream</code> accepts them, then <code>parseYaml</code> succeeds with the same result. |
| <code>parseYamlRaw_pipeline</code>     | Composition  | Raw pipeline variant without schema resolution.                                                                                                                                                    |
| <code>parseYamlRaw_ok_decompose</code> | Composition  | Every successful <code>parseYamlRaw</code> result decomposes into a successful scan step followed by a successful parse step.                                                                      |
| <code>parseYaml_ok_iff</code>          | Completeness | <code>parseYaml</code> succeeds if and only if the input is valid YAML — the bridge between the implementation and the specification.                                                              |

### **3.2.1.1. parseYaml\_pipeline**

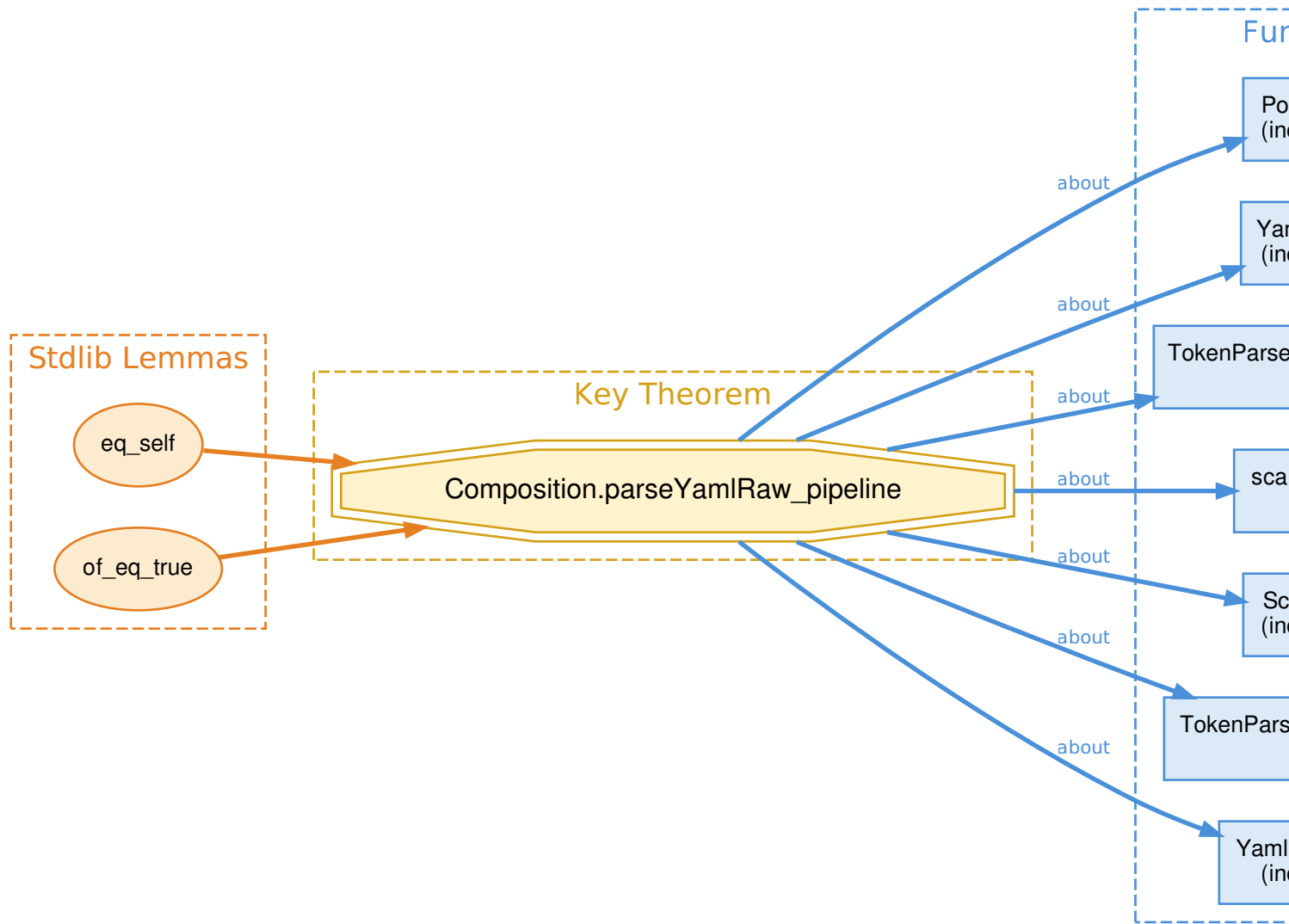


Composition.parseYaml\_pipeline  
End-to-end scanner-parser pipeline composition  
(L4YAML.Proofs.Composition)



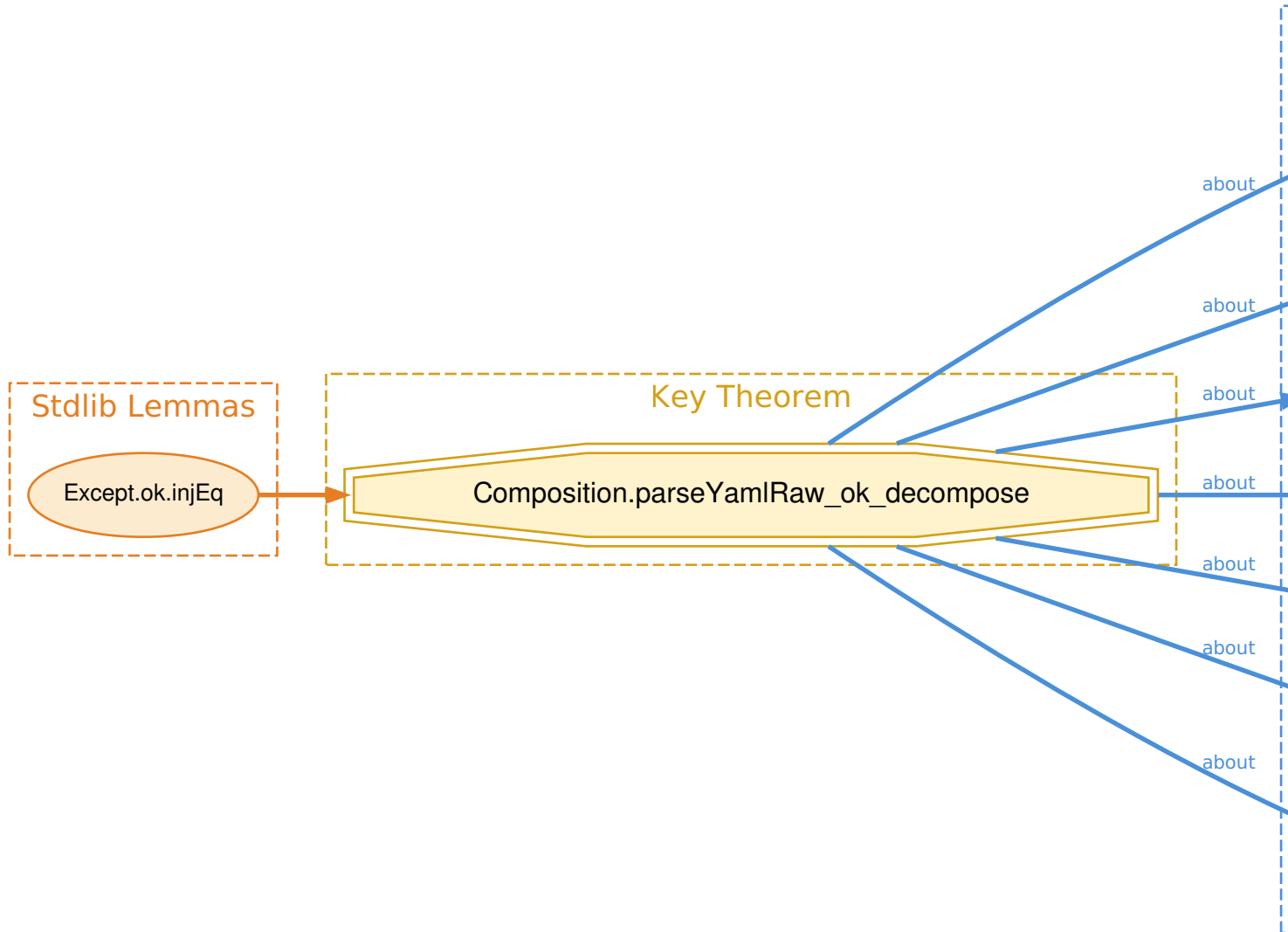
3.2.1.2. parseYamlRaw\_pipeline

Composition.parseYamlRaw\_pipeline  
Raw pipeline: scan then parse composes correctly  
(L4YAML.Proofs.Composition)



### 3.2.1.3. `parseYamlRaw_ok_decompose`

Composition.parseYamlRaw\_ok\_decompose  
Successful parse decomposes into scan + parse steps  
(L4YAML.Proofs.Composition)



### 3.2.1.4. `parseYaml_ok_iff`

`parseYaml_ok_iff` dependency graph

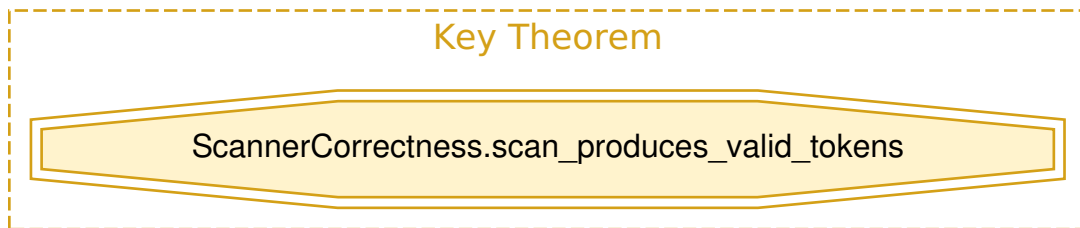
## 3.2.2. Scanner Correctness

Properties of the character-to-token scanner:

| Theorem                          | Module             | Statement                                                                                                                                                                          |
|----------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_produces_valid_tokens       | ScannerCorrectness | The scanner output satisfies ValidTokenStream: every emitted token is well-formed, positions are monotonically increasing, and the stream is bracketed by STREAM_START/STREAM_END. |
| advance_offset_lt                | ScannerProgress    | Scanner advance <i>strictly</i> increases the byte offset when the offset is within bounds — this is the core termination lemma.                                                   |
| scanLoop_success_emits_streamEnd | ScannerCorrectness | A successful scan loop always terminates with a STREAM_END token.                                                                                                                  |

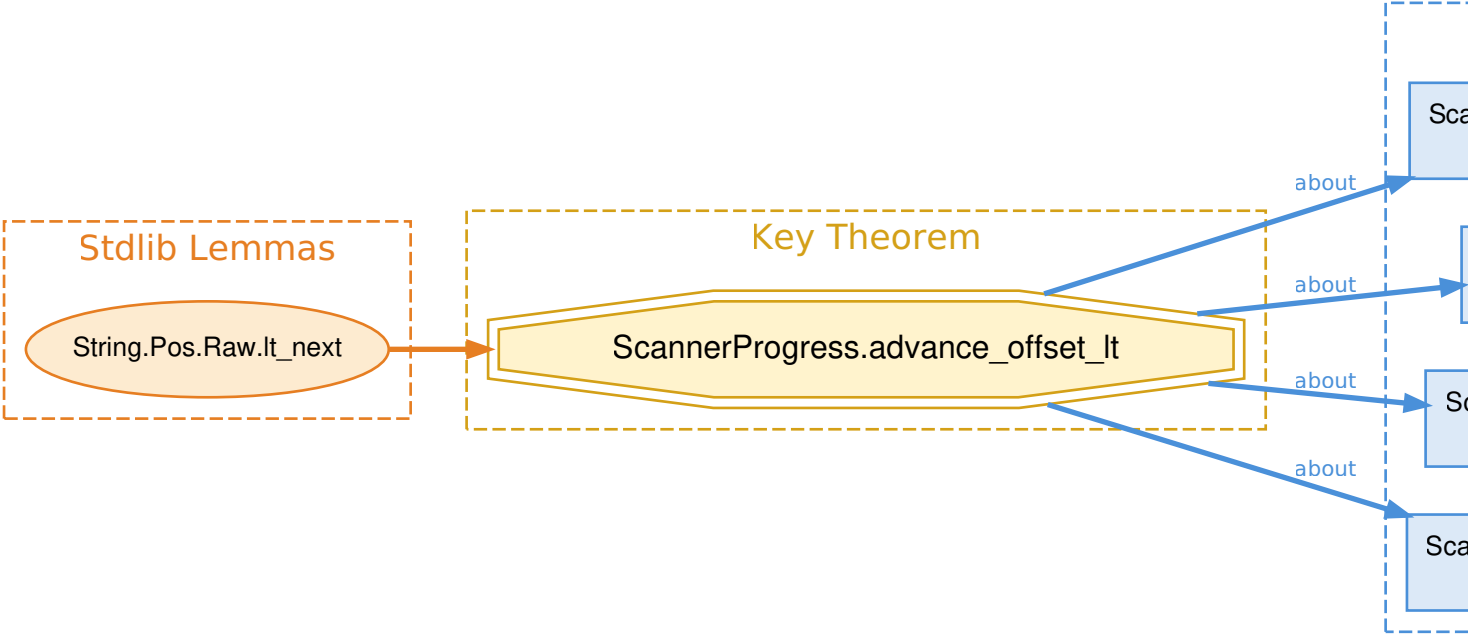
### 3.2.2.1. scan\_produces\_valid\_tokens

ScannerCorrectness.scan\_produces\_valid\_tokens  
Scanner output satisfies ValidTokenStream  
(L4YAML.Proofs.ScannerCorrectness)



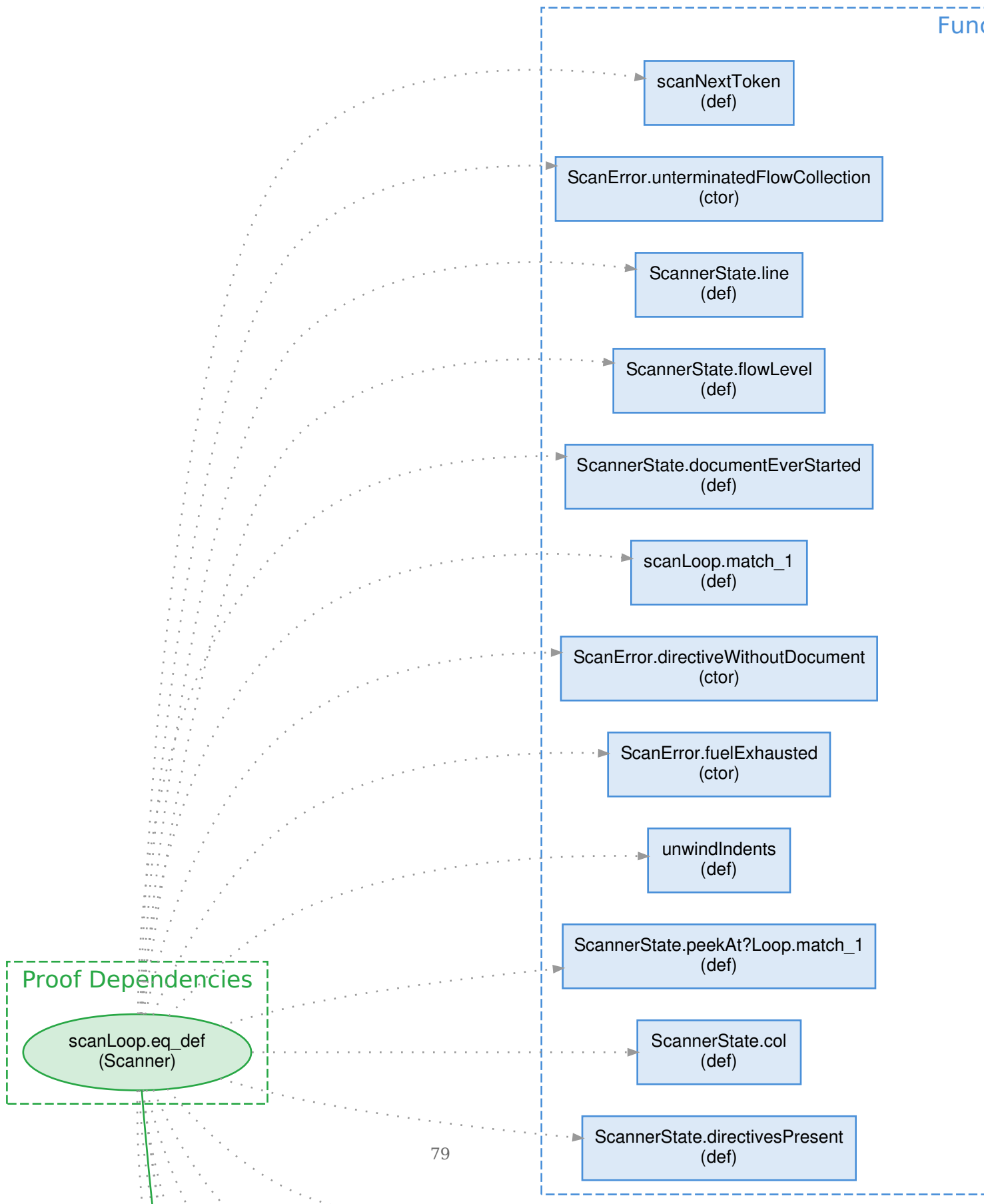
3.2.2.2. advance\_offset\_lt

ScannerProgress.advance\_offset lt  
Scanner advance strictly increases offset (termination)  
(L4YAML.Proofs.ScannerProgress)



### 3.2.2.3. scanLoop\_success\_emits\_streamEnd

ScannerCorrectness.scanLoop\_success\_emits\_stream  
Successful scan loop always emits STREAM\_END  
(L4YAML.Proofs.ScannerCorrectness)



## 3.2.3. Parser Correctness

Properties of the token-to-AST parser:

| Theorem                                            | Module                        | Statement                                                                                                                                                 |
|----------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>parseStream_sound</code>                     | <code>ParserSoundness</code>  | If the parser produces an AST, it corresponds to a valid YAML grammar derivation.                                                                         |
| <code>parseNode_anchors_grow</code>                | <code>ParserNodeProofs</code> | The anchor set grows monotonically through <code>parseNode</code> — anchors are never lost during parsing.                                                |
| <code>parseNode_aliases_resolve'</code>            | <code>ParserNodeProofs</code> | Every alias ( <code>*name</code> ) in the output of <code>parseNode</code> resolves to a previously defined anchor ( <code>&amp;name</code> ).            |
| <code>parseStream_output_anchors_wellformed</code> | <code>ParserWfaProofs</code>  | After <code>parseStream</code> completes, all output anchors are well-formed: every alias target exists and every anchor body is <code>Grammable</code> . |

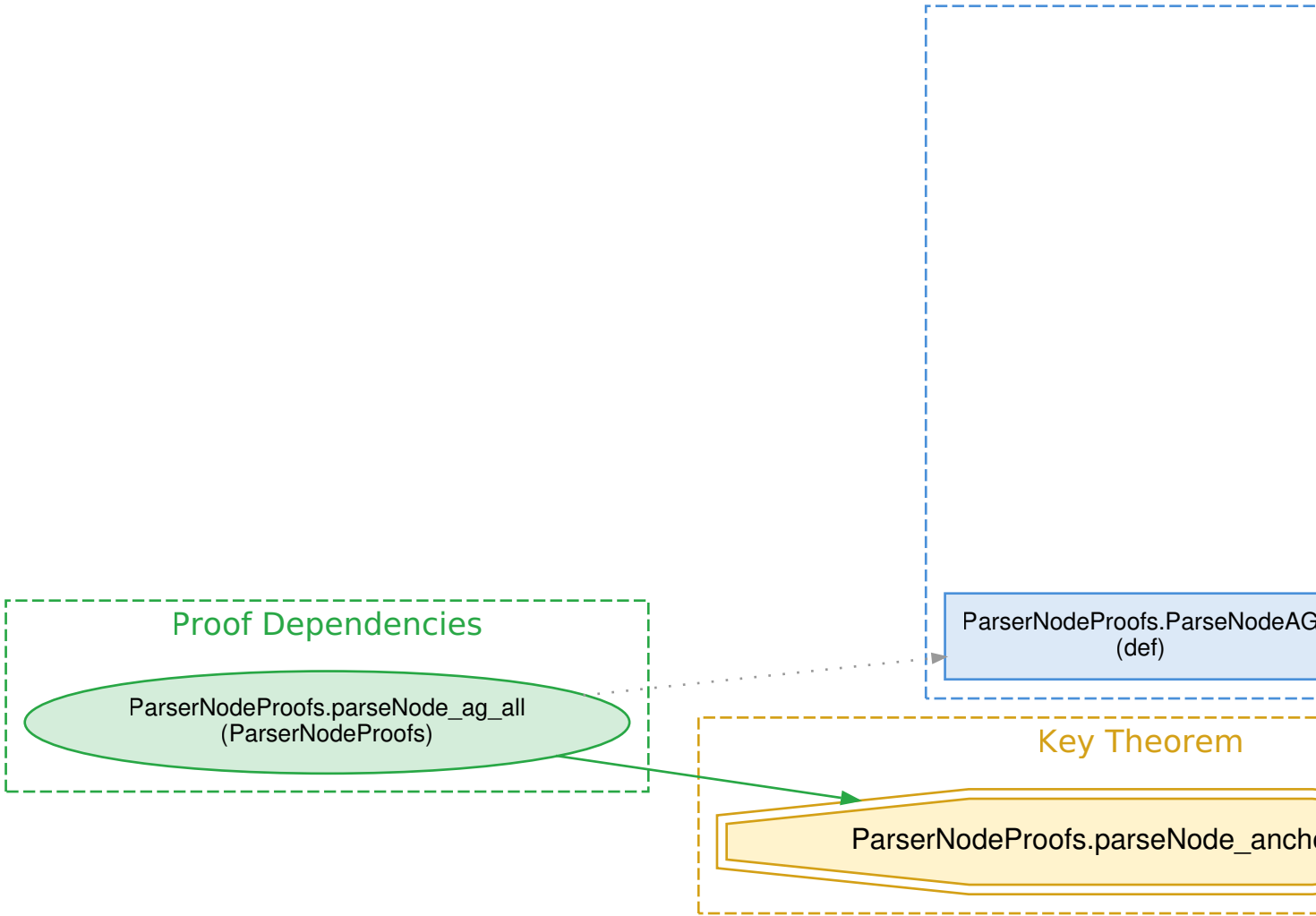
### 3.2.3.1. `parseStream_sound`

`parseStream_sound` dependency graph



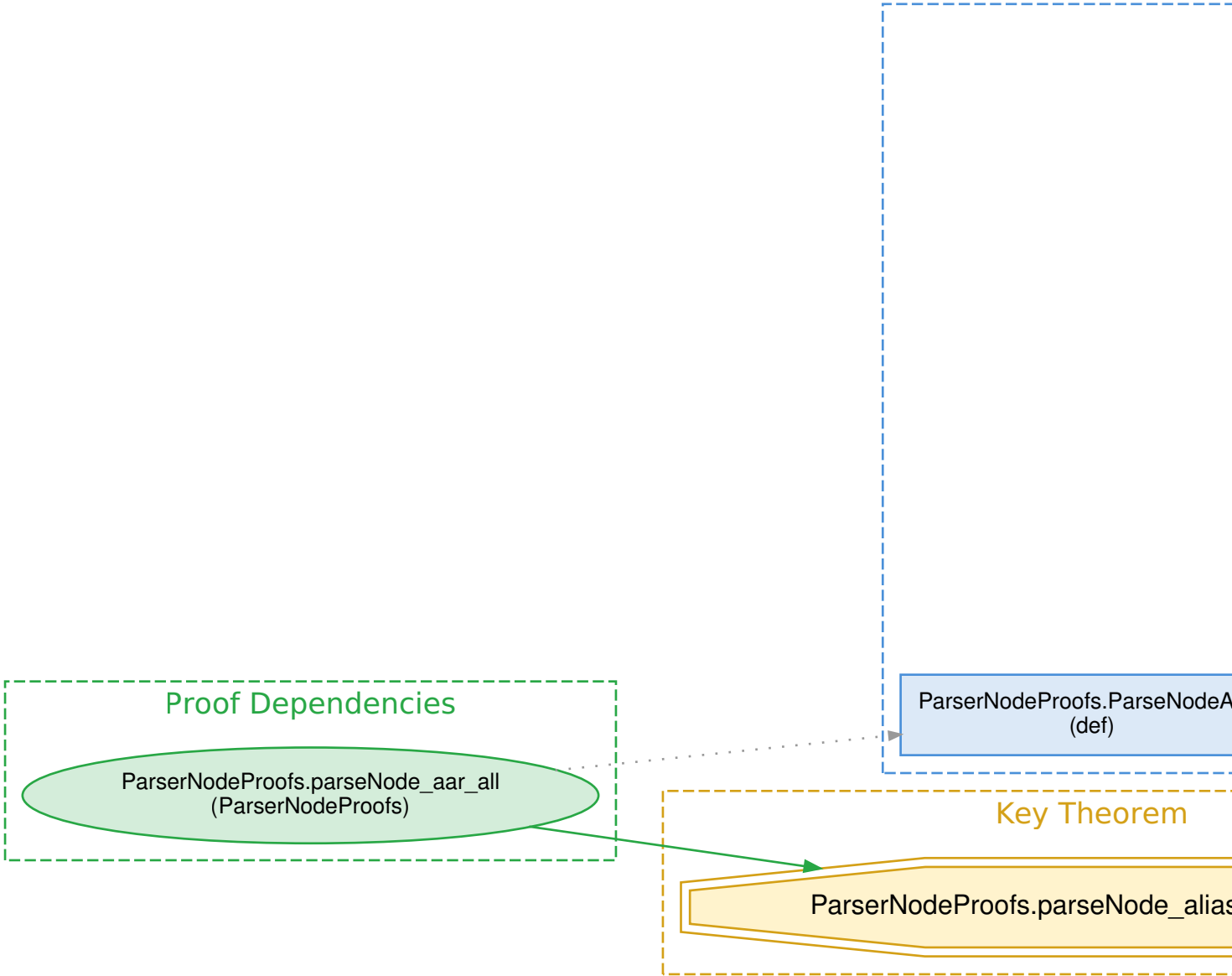
3.2.3.2. parseNode\_anchors\_grow

ParserNodeProofs.parseNode\_anchors  
Anchor set grows monotonically through p  
(L4YAML.Proofs.ParserNodeProofs



3.2.3.3. `parseNode_aliases_resolve'`

ParserNodeProofs.parseNode\_aliases  
All aliases in parseNode output resolve  
(L4YAML.Proofs.ParserNodeProofs)



#### **3.2.3.4. parseStream\_output\_anchors\_wellformed**

ParserGramm  
Output a

YamlValue  
(inductive)

Proof Dependencies

ParserGrammable.WFA\_empty  
(ParserWfaProofs)

ParserGrammable.expect\_tokens\_wfa  
(ParserWfaProofs)

ParserGrammable.parseStreamLoop\_wfa  
(ParserWfaProofs)

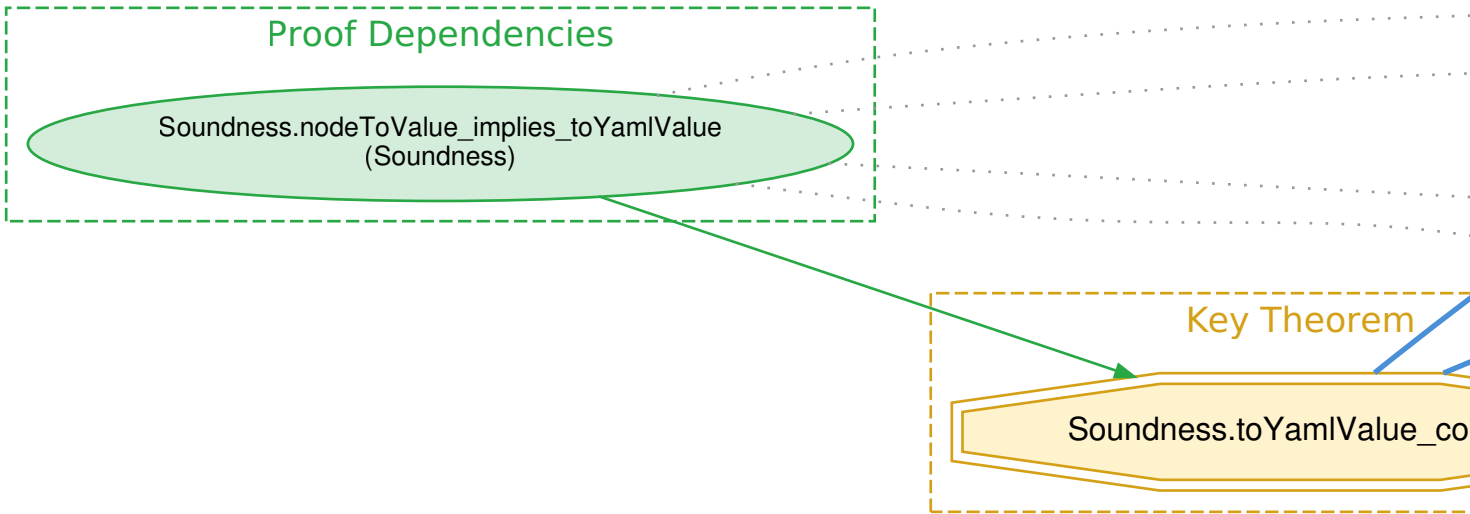
# 3.2.4. Soundness

Theorems establishing that the AST-to-value conversion faithfully implements the YAML specification:

| Theorem                   | Module    | Statement                                                                                                         |
|---------------------------|-----------|-------------------------------------------------------------------------------------------------------------------|
| toYamlValue_correct       | Soundness | The toYamlValue function correctly implements the specification's construction rules.                             |
| nodeToValue_total         | Soundness | Every well-formed AST node can be converted to a YamlValue — the conversion is total.                             |
| nodeToValue_deterministic | Soundness | AST-to-value conversion is deterministic: the same input always produces the same output.                         |
| scalar_content_preserved  | Soundness | Scalar string content is preserved through the parsing pipeline — no characters are added, dropped, or reordered. |

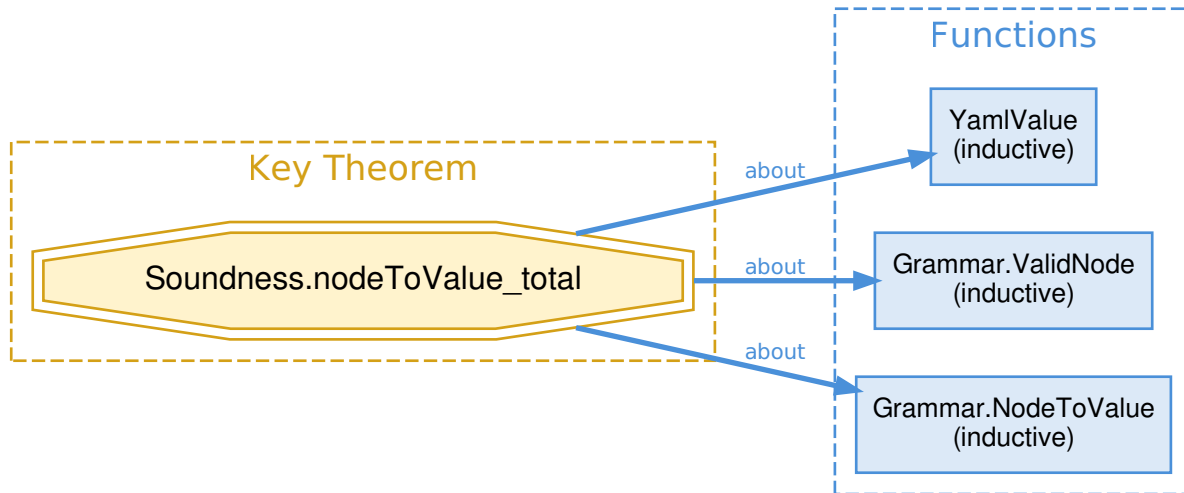
## 3.2.4.1. toYamlValue\_correct

Soundness.toYamlValue\_correct  
AST-to-value conversion matches specification  
(L4YAML.Proofs.Soundness)



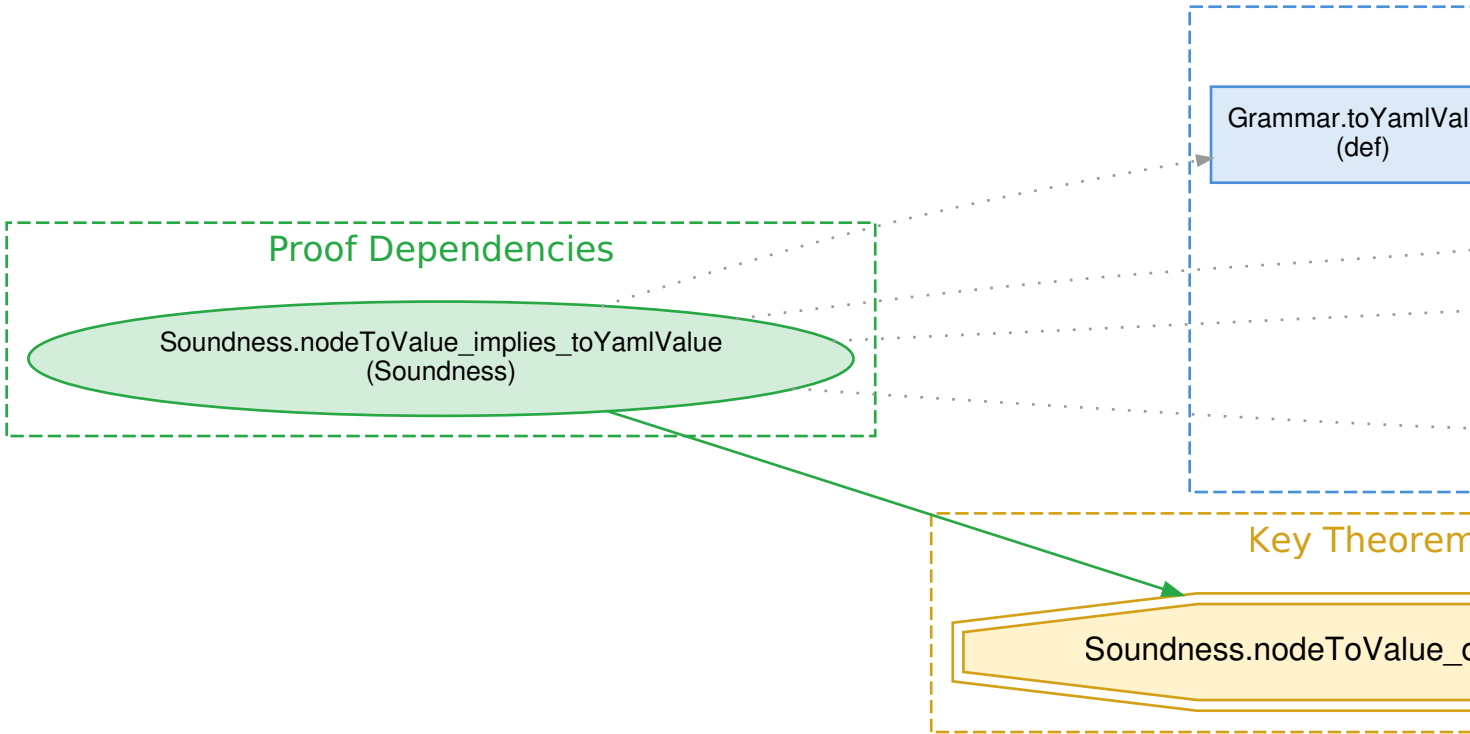
### 3.2.4.2. `nodeToValue_total`

Soundness.`nodeToValue_total`  
Every AST node can be converted to a value  
(L4YAML.Proofs.Soundness)



3.2.4.3. nodeToValue\_deterministic

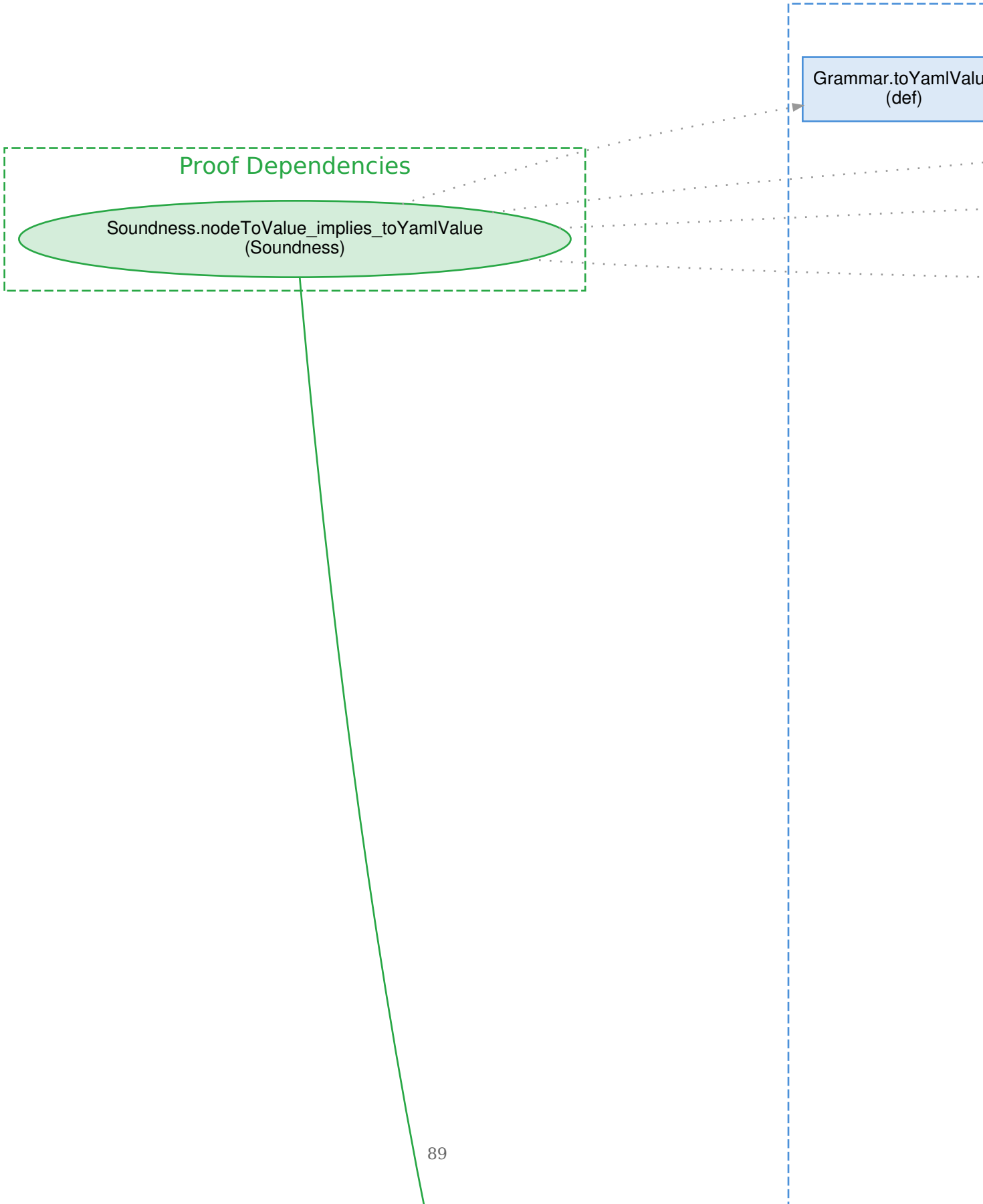
Soundness.nodeToValue\_deterministic  
AST-to-value conversion is deterministic  
(L4YAML.Proofs.Soundness)



#### **3.2.4.4. scalar\_content\_preserved**



Soundness.scalar\_content\_preserved  
Scalar content is preserved through  
(L4YAML.Proofs.Soundness)



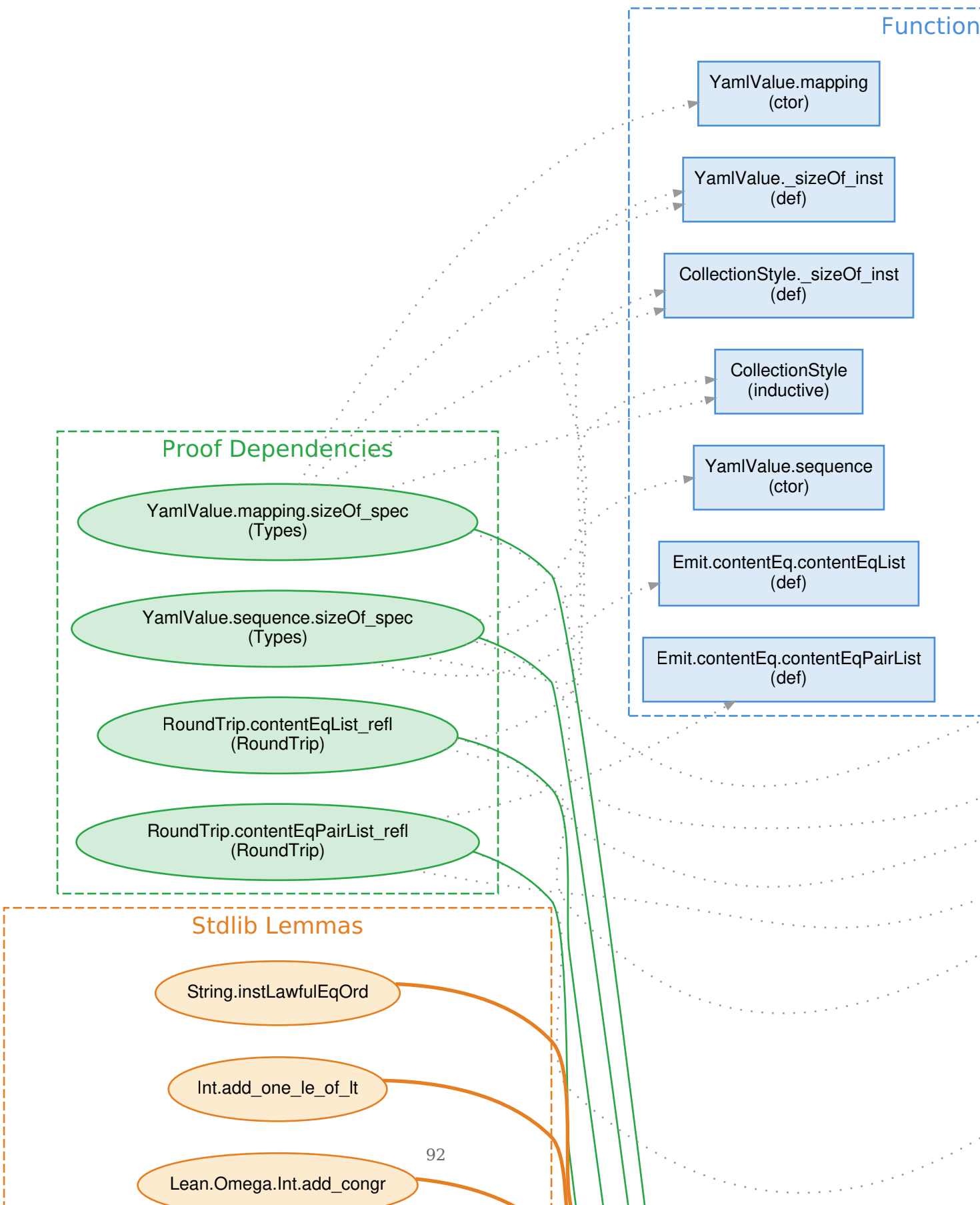
## 3.2.5. Round-Trip Properties

Theorems about the parse → emit → parse cycle:

| Theorem                | Module            | Statement                                                                                                              |
|------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------|
| contentEq_refl         | RoundTrip         | Content equality is reflexive: every YAML value is content-equal to itself.                                            |
| contentEq_symm         | RoundTrip         | Content equality is symmetric.                                                                                         |
| contentEq_trans        | RoundTrip         | Content equality is transitive — together with reflexivity and symmetry, this makes contentEq an equivalence relation. |
| emit_content_invariant | ScannerEmitBridge | The emitter preserves content equality: if two values are content-equal, their emitted canonical forms are identical.  |
| escapeTag_roundtrip    | RoundTrip         | Tag URI escape and unescape are inverse operations.                                                                    |

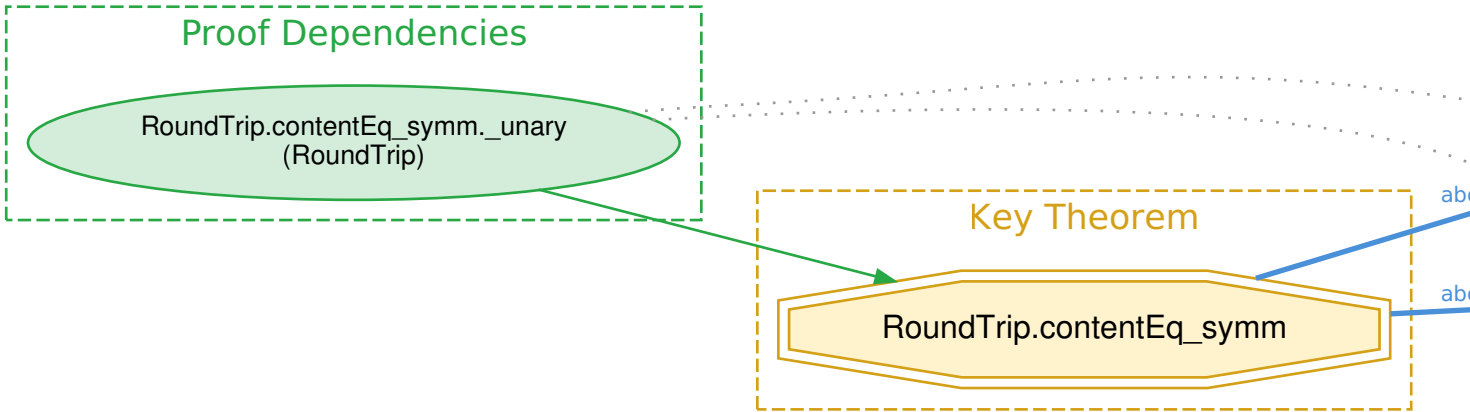
### 3.2.5.1. contentEq\_refl

RoundTrip.contentEq\_refl  
Content equality is reflexive  
(L4YAML.Proofs.RoundTrip)



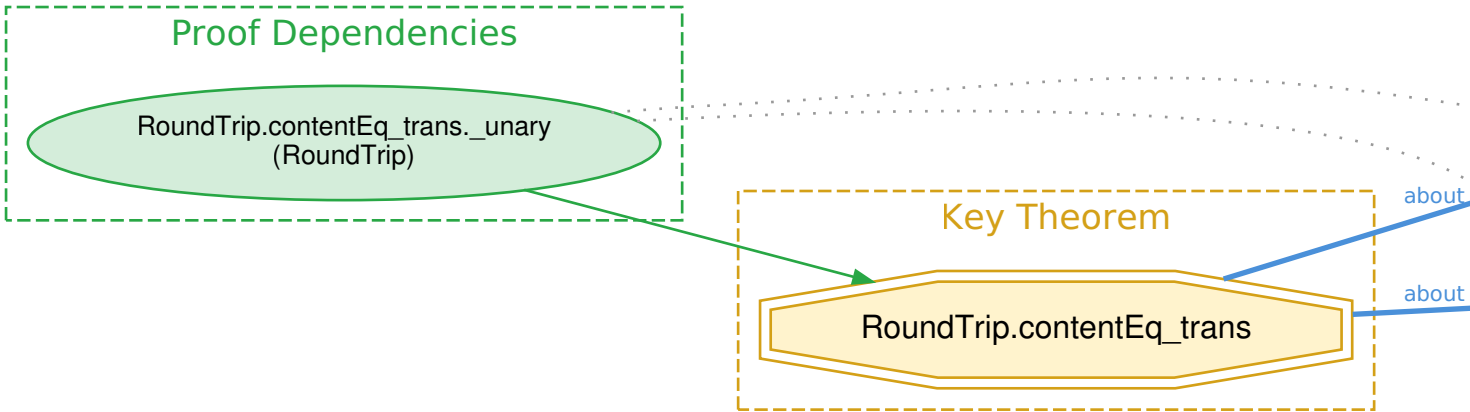
3.2.5.2. contentEq\_symm

RoundTrip.contentEq\_symm  
Content equality is symmetric  
(L4YAML.Proofs.RoundTrip)



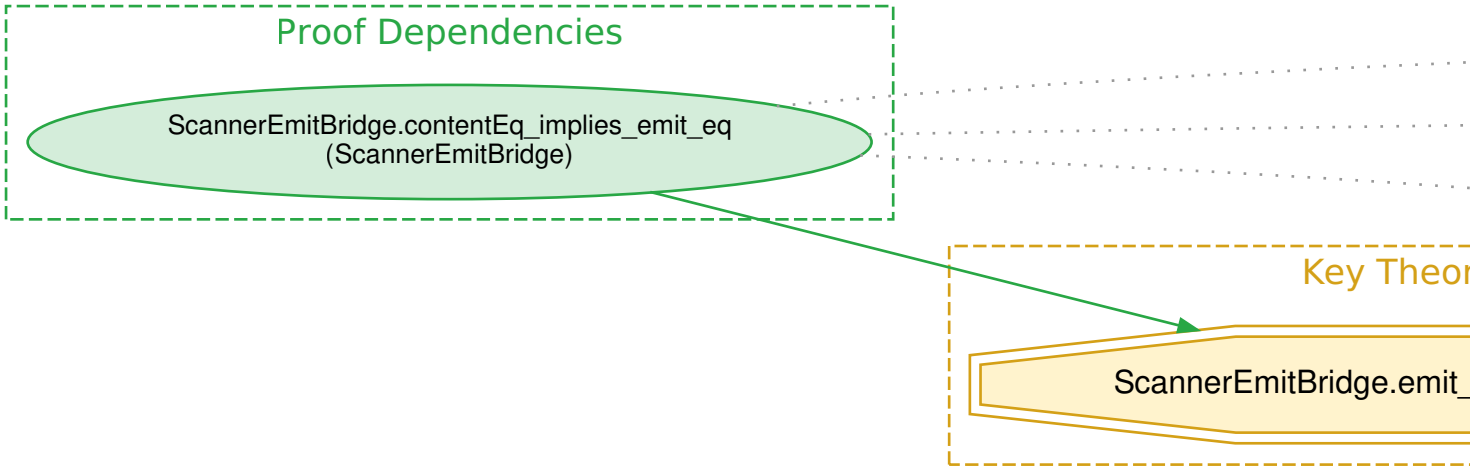
3.2.5.3. contentEq\_trans

RoundTrip.contentEq\_trans  
Content equality is transitive  
(L4YAML.Proofs.RoundTrip)



3.2.5.4. emit\_content\_invariant

ScannerEmitBridge.emit\_content\_invariant  
Emitter preserves content equality  
(L4YAML.Proofs.ScannerEmitBridge)



### **3.2.5.5. escapeTag\_roundtrip**

## Stdlib Lemmas

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_2`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_10`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_3`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_12`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_4`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_5`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_8`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_7`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_11`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_6`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_1`

`_private.L4YAML.Proofs.RoundTrip.0.L4YAML.Proofs.RoundTrip.isEscapedChar.match_1.eq_9`





## 3.5. Proof Engineering Patterns

Several patterns emerged during the verification effort:

- *Decomposition* — large functions are decomposed into validation (error guards), state transformation (pure updates), and emission (token output) phases, each proved independently then composed.
- *Append-only invariant* — the switch from `insertAt` to placeholder reservation slots with `setIfInBounds` backpatching eliminated the hardest class of proof obligations (index shifting).
- *Monotonic progress* — proving `offset_lt` (strict increase) for every scanner operation provides termination and guarantees no infinite loops.
- *Well-formedness threading* — a `WellFormed` predicate on scanner state is threaded through every operation, establishing that invariants are maintained from `scannerInit` through `scanNextToken` to stream completion.
- *Anchor monotonicity* — the `AnchorGrow` relation is proved transitively across all 14 mutually recursive parser functions, establishing that anchors accumulate but are never dropped.
- *Fuel-based termination* — the parser's 14 mutual functions use `fuel : Nat` as a decreasing argument. Initial fuel is set to `4 * tokens.size + 4`, large enough for any valid input.



## 3.4. Theorem Dependency Visualization

L4YAML includes tooling for visualizing theorem dependencies as bipartite graphs. Each graph shows a key theorem at the center, with:

- *Functions* (left, blue) — the implementation functions the theorem proves a property about
- *Proof dependencies* (right, green) — the supporting theorems used in the proof

### 3.4.1. Generating Graphs

The theoremgraph tool lives in the [L4YAML.FGM](https://github.jpl.nasa.gov/pass/L4YAML.FGM) (<https://github.jpl.nasa.gov/pass/L4YAML.FGM>) bridge project. Generate DOT files for all key theorems:

```
cd ../L4YAML.FGM
lake build theoremgraph
lake exe theoremgraph tmp/graphs
```

Render to SVG with Graphviz:

```
for f in tmp/graphs/*.dot; do
  dot -Tsvg "$f" -o "${f%.dot}.svg"
done
```

Then open tmp/graphs/index.html for an overview page.

Generate a single theorem's graph:

```
lake exe theoremgraph --dot parseYaml_pipeline
```

List all available key theorems:

```
lake exe theoremgraph --list
```

## 3.4.2. Dependency Extraction

The lower-level depgraph tool extracts three complete dependency graphs from the L4YAML environment:

1. **Function call graph** — definition  $\rightarrow$  definition calls
2. **Theorem dependency graph** — theorem  $\rightarrow$  theorems used in proof
3. **Function–theorem map** — which functions each theorem is about

```
lake build depgraph
lake exe depgraph --dot calls > calls.dot
lake exe depgraph --dot thmdeps > thmdeps.dot
lake exe depgraph --dot about > about.dot
lake exe depgraph --stats
```

## 3.4.3. Theorem Coverage Analysis

The analyzethms tool identifies leaf theorems (proved but never cited in other proofs) and native\_decide usage patterns:

```
lake build analyzethms
lake exe analyzethms tmp/analysis
```

This writes leaf\_thms.json, native\_decide\_leaves.json, duplicates.json, and stats.txt.



## 3.1. Three-Layer Strategy

### 3.1.1. Layer 1: Machine-Checked Proofs

The core layer consists of 2,309 Lean 4 theorems across 61 proof modules (~47,000 lines). These proofs are checked by the Lean kernel — the small trusted core of the system — and establish properties including:

- **Soundness** — every token stream produced by the scanner corresponds to a valid YAML grammar derivation
- **Completeness** — every valid YAML input is accepted (not rejected with an error)
- **Progress** — the scanner's input offset strictly increases on every step, guaranteeing termination
- **Well-formedness preservation** — internal invariants (indentation stack consistency, flow level balance, simple key lifecycle) are maintained across all scanner operations
- **Pipeline composition** — scanner and parser compose correctly to deliver end-to-end guarantees

### 3.1.2. Layer 2: Compile-Time Guards

2,124 `#guard` statements are evaluated by the Lean kernel at build time. These are not runtime tests — they are *kernel-evaluated assertions* that must hold for the project to compile. A failing `#guard` is a build error, not a test failure.

Guards are used extensively for:

- Concrete scanner behavior on specific inputs
- Round-trip properties (parse → emit → parse = original)
- Token stream structure for specification examples
- Character predicate boundary conditions

### 3.1.3. Layer 3: Runtime Tests

1,041 runtime tests across 19 suites provide additional coverage:

- *Specification examples* — all 132/132 YAML 1.2.2 examples

- *yaml-test-suite* — 225/225 applicable test IDs (354/406 total; 52 YAML 1.1/1.3 tests are correctly skipped)
- *Property tests* — randomized input generation for edge cases
- *Mutation tests* — systematic input perturbation
- *Adversarial tests* — handcrafted inputs targeting parser limits
- *Round-trip tests* — parse → dump → parse cycle validation





## 3.6. Zero-Axiom Policy

The project uses zero axioms beyond Lean's built-in foundations (`propext`, `Quot.sound`, `Classical.choice`). No `sorry` appears anywhere in the codebase. No `partial def` is used — every function has a kernel-checked termination proof.

This means the formal guarantees are as strong as the Lean kernel itself: if the kernel accepts the proofs, the properties hold.





# Welcome to the documentation page

This was built using Lean 4 [4.30.0-rc1](https://github.com/leanprover/lean4/tree/714601baf118066cbf3f282361339c6d06665b2a) (https://github.com/leanprover/lean4/tree/714601baf118066cbf3f282361339c6d06665b2a)

I













# lean4-yaml-verified Test Coverage

YAML 1.2.2 verified parser · Lean 4

## yaml-test-suite Compliance

```
Total unique tests:      406
Applicable (YAML 1.2.2): 358 (48 skipped – YAML 1.3 specific)
Passed:                  263
Failed:                   0
Expected fail:           95
Unexpected pass:          0
Timeout:                  0

Correct: 358/358 (100.0%)
```

## Reports

[Full Coverage Report](#)

All 406 unique test cases with filtering & sorting

## Coverage by Stage

[scalar](#)

58/58 correct (100.0%) · 24 YAML 1.3 skipped

[flow](#)

46/46 correct (100.0%)

[block](#)

99/99 correct (100.0%)

[document](#)

17/17 correct (100.0%) · 7 YAML 1.3 skipped

[advanced](#)

64/64 correct (100.0%) · 17 YAML 1.3 skipped

[error](#)

74/74 correct (100.0%)

# Suite Runner (Progressive Stages)

✓ Suite Runner: 869/869 (100.0%)

Each stage tests its own tests plus all lower stages (cumulative). Error tests excluded.

| Stage    | Total | Passed | Failed | Skipped |
|----------|-------|--------|--------|---------|
| scalar   | 82    | 58     | 0      | 24      |
| flow     | 128   | 104    | 0      | 24      |
| block    | 227   | 203    | 0      | 24      |
| document | 251   | 220    | 0      | 31      |
| advanced | 332   | 284    | 0      | 48      |
| Total    | 1020  | 869    | 0      | 151     |

**Verified Tests**

**X Internal Verified Test Suites (2305/2323)**

|                                                                                                                                                                                                                                                                              |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Unit Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/Main.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/Main.lean</a> )                                                                            | ✓ 10/10        |
| <b>Explicit Key Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ExplicitKeyTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ExplicitKeyTests.lean</a> )                                            | ✓ 149/149      |
| <b>Flow Completeness Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/FlowTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/FlowTests.lean</a> )                                                     | ✓ 88/88        |
| <b>Structural Validation Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ValidationTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ValidationTests.lean</a> )                                     | ✓ 84/84        |
| <b>Dump Round-Trip Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/DumpRoundTrip.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/DumpRoundTrip.lean</a> )                                               | ✓ 117/117      |
| <b>Raw Parse / Compose Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/RawParseTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/RawParseTests.lean</a> )                                           | ✓ 29/29        |
| <b>YAML 1.2.2 Spec Examples</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SpecExamples.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SpecExamples.lean</a> )                                              | ✓ 132/132      |
| <b>Schema ↔ Dump Integration</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SchemaDump.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SchemaDump.lean</a> )                                                 | ✓ 68/68        |
| <b>Scanner &amp; TokenParser Tests (Phase 9)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerTests.lean</a> )                             | ✓ 32/32        |
| <b>YAML 1.2.2 Spec Examples (Scanner/Parser Pipeline)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerSpecExamples.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerSpecExamples.lean</a> )      | ✓ 132/132      |
| <b>Adversarial Grammar-Directed Tests (v0.2.13.1)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/AdversarialGrammarTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/AdversarialGrammarTests.lean</a> )  | ✓ 154/154      |
| <b>Adversarial Instantiation Tests (sorry audit)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/AdversarialInstantiation.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/AdversarialInstantiation.lean</a> ) | ✗<br>1073/1091 |
| <b>Mutation Testing on yaml-test-suite (v0.2.13.2)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/MutationSuiteTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/MutationSuiteTests.lean</a> )           | ✓ 45/45        |
| <b>Property-Based Round-Trip Tests (v0.2.13.3)</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/PropertyTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/PropertyTests.lean</a> )                         | ✓ 124/124      |
| <b>Production Coverage Analysis</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ProductionCoverage.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ProductionCoverage.lean</a> )                              | ✓ 18/18        |
| <b>Parser Security Limit Tests</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/LimitTests.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/LimitTests.lean</a> )                                               | ✓ 43/43        |
| <b>Demo</b> ( <a href="https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Demo.lean">https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Demo.lean</a> )                                                                                              | ✓ 7/7          |

|       |           |
|-------|-----------|
| Total | 2305/2323 |
|-------|-----------|

[Verified Tests Detail](#)

2305/2323 tests across 17 suites

[YAML 1.2.2 Production Coverage](#)

All productions cross-referenced with @[yaml\_spec] annotations

Generated by [yaml-test-suite](https://github.com/yaml/yaml-test-suite) (https://github.com/yaml/yaml-test-suite) runner · Lean 4